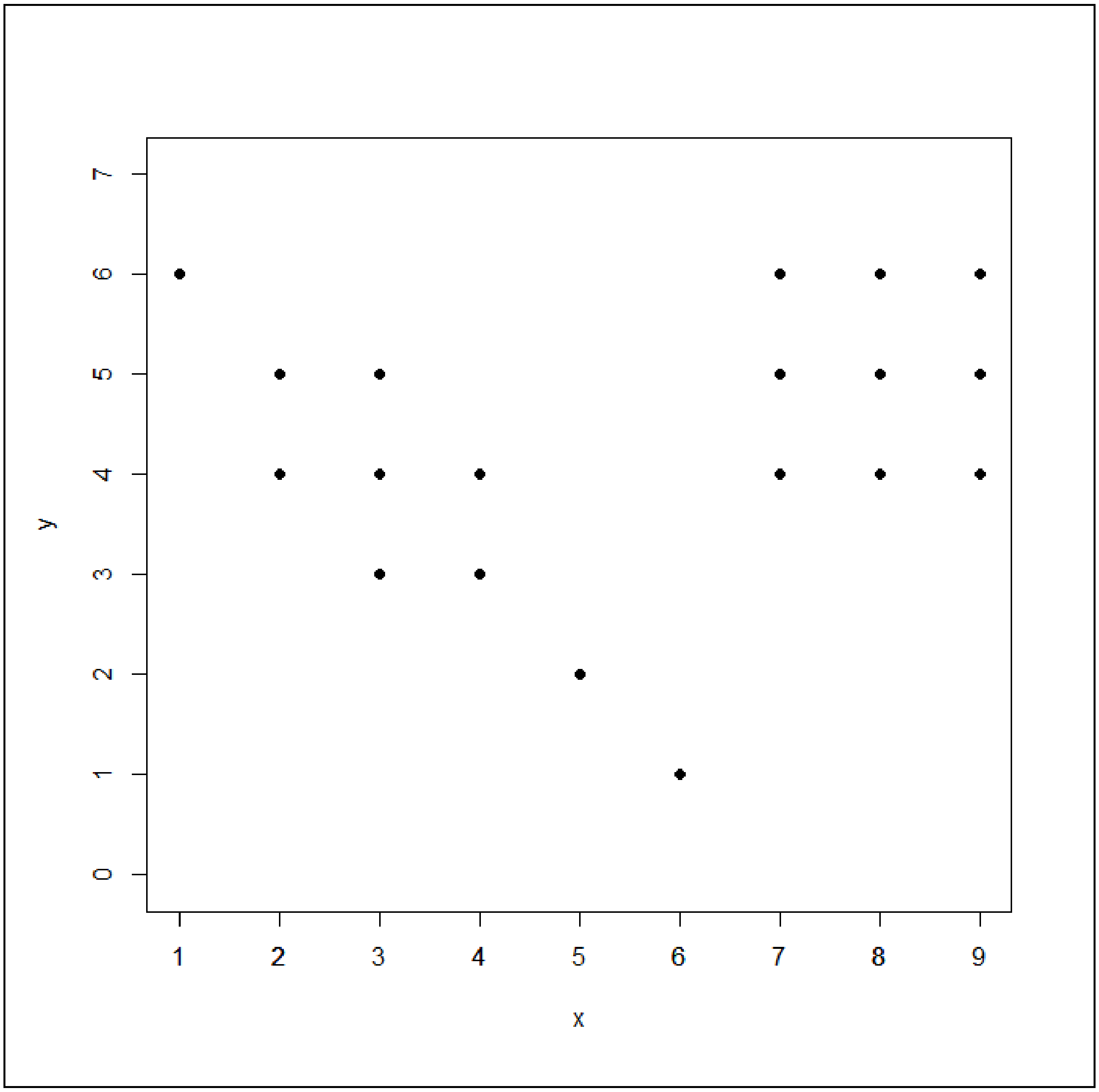
WS 23 Understanding DBSCAN

Study the output shown and answer the questions that follow.

```
x<-c(1,2,2,3,3,3,4,4,5,6,7,7,7,8,8,8,9,9,9)
y<-c(6,5,4,5,4,3,4,3,2,1,6,5,4,6,5,4,6,5,4)
plot(x,y)
axis(1, at = seq(1, 9, by=1))
points(x, y, pch = 19)
```



(a) How many observations (points) are shown in the plot above?

(b) Suppose we use DBSCAN with eps = 1 and MinPts = 3 to cluster the data (remember: MinPts include the point itself). Beside each observation, write the letter C if the point is a core point, B if the point is a border point and O if it is an outlier.

(c) How many clusters will DBSCAN unravel?

```
library(fpc)
library(dbscan)
library(factoextra)
df<-data.frame(x,y)
set.seed(123)
db<-fpc::dbscan(df, eps=1, MinPts = 3)
fviz_cluster(db, data =df, stand=FALSE, ellipse=FALSE, show.clust.cent=FALSE, geom="point",
palette="jco", ggtheme= theme_classic())
#diagram not shown as the colors identifying cluster(s) and outliers will not show well on this black
and white printout.  Please try out the code above on your laptop/PC.
print(db)
```

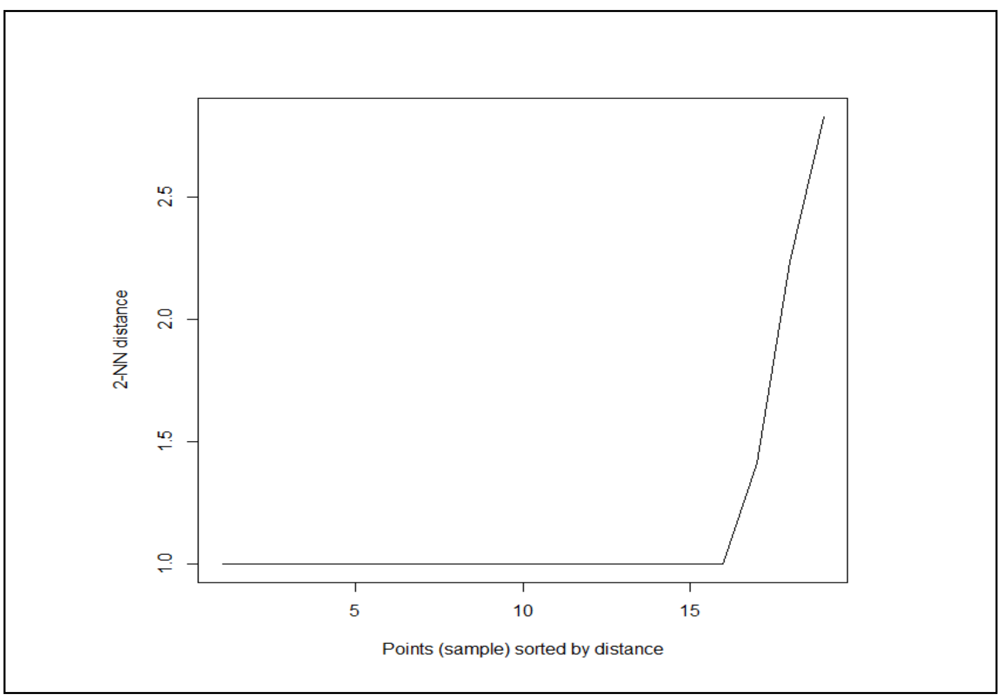
(d) Complete the print(db) output below. You will need to draw more boxes if there are more than 1 cluster.

dbscan Pts=xx MinPts=3 eps=1

	0	1
border		
seed		
total		

(e)

```
dbscan::kNNdistplot(df, k = 2)
```



Calculate the largest 2-NN distance accurate to 4 decimal places.

(f) Write down two advantages of using DBSCAN.

Not shown in textbook, but you can obtain the k-NN distances by this command:

```
dbscan::kNNdist(df, k = 2)
 [1] 2.236068 1.000000 1.000000 1.000000 1.000000 1.000000 1.000000 1.000000
 [9] 1.414214 2.828427 1.000000 1.000000 1.000000 1.000000 1.000000 1.000000
[17] 1.000000 1.000000 1.000000
> sqrt(8)
[1] 2.828427
```